

Domain for z : All the courses in the school

$$\exists x \exists y \forall z (T(x, y, z))$$

f) Some student in this class grew up in the same town as exactly one other student in this class

$G(x, y)$: Student x grew up in Town y .

Domain of x : All students of the class

Domain of y : All Towns in the world

$$\exists x \exists y (G(x, y) \wedge \exists z ((x \neq z) \wedge G(z, y)) \wedge \forall w ((w \neq x) \wedge G(w, y) \Rightarrow w = z)))$$

g) Every student in this class has chatted with at least one other student in at least one chat group:

$G(x, y, z)$: Student x has chatted with student y in chat group z .

Domain of x and y : All students of the class

Domain of z : All chat groups

$$\forall x \exists y \exists z ((x \neq y) \wedge G(x, y, z))$$

15) Use quantifiers and predicates with more than one variable to express these statements. (Common domain for x : All students of the class)

a) Every computer science student needs a course in discrete mathematics

$D(x, y)$: Person x is a student of department y .

$C(x, y)$: Person x needs course y .

Domain for y are respectively the set of all departments and the set of all courses.

$$\forall x (D(x, \text{Computer Science}) \Rightarrow C(x, \text{Discrete Mathematics}))$$

b) There is a student in this class who owns a personal computer

$O(x, y)$: Student x owns personal computer y .

Domain of y : All personal computers in the world

$$\exists x \exists y O(x, y)$$